

10(a)	heating depends on current ² / I^2	B1
	and current ² / I^2 is always positive	B1
	or	
	a.c. changes direction (every half cycle)	(B1)
	but heating effect is independent of current direction	(B1)
	or	
	voltage and current are always in phase in a resistor	(B1)
	so $V \times I$ is always positive	(B1)
	or	
	sketch graph drawn showing power against time	(B1)
	comment that power is always positive	(B1)
10(b)(i)	for same power (transmission, higher voltage) → lower current	B1
	lower current → less power loss in (transmission) cables	B1
10(b)(ii)	- voltage can be (easily) stepped up/down - transformers only work with a.c. - generators produce a.c. - easier to rectify than invert <i>Two sensible suggestions, 1 mark each.</i>	B2